The word-enumerator formula for every arbor polytope

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Abstract

We prove Conjecture 5.2 of Athanasiadis, Xiao, and Yan for every arbor, including arbitrary branching and nonsingleton vertex blocks. The result is a direct corollary of the preorder descent theorem proved in our companion MathDB 374958 solution. We give the precise arbor reduction and reproduce the reciprocity and word-counting proof in full.

Task 1. Full solution

An arbor τ on $[n]$ is a rooted tree whose vertices are the nonempty blocks of a partition of $[n]$, where $n \geq 1$. Let $D(v)$ be the union of the blocks in the subtree rooted at v, including its own block. The arbor polytope is

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}^n_{\geq 0} : x(D(v)) \leq |D(v)| \text{ for every vertex } v\}, \qquad x(A) = \sum_{i \in A} x_i.$$

Let W_τ consist of words $w \in [n]^n$ in which the letters from $D(v)$ occur at least $|D(v)|$ times for every v. Let $\mathrm{des}(w) = \#\{j \in [n-1] : w_j > w_{j+1}\}$.

Theorem 1. *For every arbor τ of size n,*

$$\boxed{h^*(\mathcal{Q}_\tau, t) = \sum_{w \in W_\tau} t^{n-1-\mathrm{des}(w)}.}$$

From descendant sets to order ideals

Define a preorder σ on $[n]$ by $i \leq_\sigma j$ when the block of i is a descendant of the block of j. Elements in one block are equivalent. Thus $D(v)$ is a principal order ideal. Every order ideal I of σ is the disjoint union of the sets $D(v)$ for its maximal blocks v. The subtrees are disjoint because these blocks are incomparable in a rooted tree.

Summing over this disjoint union shows that the upper bounds on $D(v)$ are equivalent to $x(I) \leq |I|$ for every ideal I. It follows that $\mathcal{Q}_\tau = \mathcal{Q}_\sigma$, where $\mathcal{Q}_\sigma$ denotes the preorder polytope. The identical argument with lower bounds on letter multiplicities gives $W_\tau = W_\sigma$, where W_σ uses all order ideals. This reduction respects the usual order on the alphabet, which need not be compatible with σ.

For completeness we prove the needed preorder formula below. Set $E = [n]$. Write σ^* for the reversed preorder and $P_{\sigma^*} = \mathcal{Q}_{\sigma^*} \cap \mathbb{Z}^n$.

A counting identity from reciprocity

We give the needed reciprocity argument in full. Let

$$\mathcal{Q}_\sigma(r,s) = \left\{ x \in \mathbb{R}^E_{\geq 0} : \sum_{e \in I} x_e \leq r|I| + s \text{ for every nonempty order ideal } I \right\}.$$

By [2, Propositions 3.2–3.3 and 7.1], for nonnegative integers r, s this is a lattice polytope, also expressible as an integral Minkowski sum of simplices. Its lattice-point count is the polynomial $\mathbb{E}_\sigma(r,s)$, of total degree n.

For positive integers r, s, the polytope is full dimensional (it contains $[0,r]^E$). Its interior integer points satisfy $x_e \geq 1$ and $\sum_{e\in I} x_e \leq r|I| + s - 1$ for every nonempty order ideal. Subtracting the all-ones vector therefore bijects these points with $\mathcal{Q}_\sigma(r-1, s-1) \cap \mathbb{Z}^E$. All inequality normals are nonzero; redundant inequalities have no effect on this strict-interior characterization.

For $P = \mathcal{Q}_\sigma(r,s)$ one has $kP = \mathcal{Q}_\sigma(kr, ks)$ for nonnegative integers k, so its Ehrhart polynomial equals $\mathbb{E}_\sigma(kr, ks)$ as a polynomial in k. Ordinary Ehrhart reciprocity at $k = -1$ and the translation just described imply

$$\mathbb{E}_\sigma(-u, -v) = (-1)^n \mathbb{E}_\sigma(u-1, v-1). \tag{1}$$

Initially this holds for positive integer pairs (u, v); it is a polynomial identity because a polynomial vanishing on that entire grid is zero (first fix v and vary u, then vary v).

For $a = (a_e)_{e\in E}$ write $|a| = \sum_e a_e$ and $a_0 = n - |a|$. The established formula [2, Lemma 4.1 and Equation (25)] is

$$\mathbb{E}_\sigma(u,v) = \sum_{a \in P_{\sigma^*}} \binom{v + a_0}{a_0} \prod_{e\in E} \binom{u + a_e - 1}{a_e}. \tag{2}$$

Here $a_0 \geq 0$, and binomial coefficients with fixed nonnegative lower argument are interpreted as polynomials in the upper argument. Putting $(u, v) = (-m-1, -1)$ in (2), every term with $a_0 > 0$ vanishes, since $\binom{a_0 - 1}{a_0} = 0$. For the remaining terms $|a| = n$, use

$$\binom{a_e - m - 2}{a_e} = (-1)^{a_e} \binom{m+1}{a_e}.$$

Together with (1), this gives, for all $m \geq 0$,

$$\mathrm{Ehr}(\mathcal{Q}_\sigma, m) = \mathbb{E}_\sigma(m, 0) = \sum_{\substack{a \in P_{\sigma^*} \\ |a| = n}} \prod_{e\in E} \binom{m+1}{a_e}. \tag{3}$$

This implication of bivariate reciprocity was already noted, conditionally, in [2, Remark 7.3]; equation (1) supplies the missing premise.

An elementary multiset identity

For a nonnegative integer vector a with $|a| = n$, let S_a be all words having a_e copies of each e. We prove

$$\sum_{m \geq 0} \left(\prod_e \binom{m+1}{a_e} \right) t^m = \frac{\displaystyle\sum_{w \in S_a} t^{n-1-\mathrm{des}(w)}}{(1-t)^{n+1}}. \tag{4}$$

Choose, for each letter e, a subset of a_e bins from the $m+1$ ordered bins $0, 1, \dots, m$. Place one copy of e in each chosen bin. The number of choices is the product on the left. Within each bin, read its distinct letters in strictly decreasing order, and concatenate the bins. This produces a word $w \in S_a$ and a weakly increasing sequence of bin indices $0 \le b_1 \le \cdots \le b_n \le m$.

At each index j for which $w_j \le w_{j+1}$, the bin index must strictly increase. Conversely these conditions suffice: a constant bin segment is strictly decreasing and contains no repeated letter. Thus this construction is a bijection. The number of required strict increases is $c(w) = n-1-\mathrm{des}(w)$. Subtract from b_j the number of required increases before position j. When $m \ge c(w)$ this gives a weakly increasing sequence of length n in $\{0, \dots, m-c(w)\}$, counted by $\binom{m-c(w)+n}{n}$. There are no such sequences for $m < c(w)$. Consequently the contribution of w to the generating function is

$$\sum_{m \ge c(w)} \binom{m-c(w)+n}{n} t^m = \frac{t^{c(w)}}{(1-t)^{n+1}},$$

which proves (4), including repeated letters and $m = 0$.

Proof of Theorem 1. A vector a of total sum n belongs to P_{σ^*} exactly when $\sum_{e \in J} a_e \le |J|$ for every order filter J of σ. Taking complements $I = E \setminus J$, these conditions are equivalent to $\sum_{e \in I} a_e \ge |I|$ for every order ideal I of σ. Therefore the sets S_a indexed in (3) partition W_σ. Sum (4) over those vectors and use (3). The definition

$$\sum_{m \ge 0} \mathrm{Ehr}(\mathcal{Q}_\sigma, m) t^m = \frac{h^*(\mathcal{Q}_\sigma, t)}{(1-t)^{n+1}}$$

now yields the desired identity. □

Verification and attribution

The conjecture and the octopus special case are due to Athanasiadis, Xiao, and Yan [1]. The preorder counting formula and the conditional deduction from bivariate reciprocity are due to Athanasiadis and Chapoton [2]. The complete reciprocity proof above is reproduced from our MathDB 374965 and 374958 packages; this separately listed arbor target is a corollary, not an independent new general theorem. Since $\mathcal{Q}_\tau = \mathcal{Q}_\sigma$ and $W_\tau = W_\sigma$, the formula proved above is exactly the target.

All 14 pages of the current arbor primary and all 26 pages of the preorder paper were read. The verifier checks every one of the 1,674 labeled arbors through size five, including arbitrary blocks: 11,275 ideal decompositions, 1,674 complete h^*-identities, 9,866 binomial counts, and 740 direct lattice enumerations. It also checks the published octopus formula and six-element examples. Bounded checks found no earlier external full target claim or identified target journal final. Review is by the authoring agent only.

Tasks 2–5. Proposed follow-up directions

Task 2. Construct a recursive bijection from dilated arbor lattice points to the bin assignments, using the branching of the rooted tree.

Task 3. Refine the formula by descent sets and derive shuffle identities for the subtrees attached to a vertex.

Task 4. Find a parking interpretation of the magic-basis coefficients for arbitrary arbors, extending the known octopus case.

Task 5. Determine which modifications of the subtree capacities admit an analogous word formula with a fixed multiplicity constraint.

References

[1] C. A. Athanasiadis, Q. Xiao, and X. Yan, *Lattice point enumeration of some arbor polytopes*, arXiv:2603.11654v1, 2026.

[2] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.